

CO4 AT02 CODING GOLF COMPETITION

QUESTIONS

Q1

Merge Sort
100 marks

1/1 answered

Q1 Merge Sort

100 marks

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

Your Code

```
1 #include<stdio.h>
2 void merge(int a[],int l,int m,int h){
3     int i=l,j=m+1,k=0,b[20];
4     while(i<=m && j<=h)
5         b[k++]=a[i]<a[j]?a[i++]:a[j++];
6     while(i<=m) b[k++]=a[i++];
7     while(j<=h) b[k++]=a[j++];
8     for(i=l,k=0;i<=h;i++,k++)
9         a[i]=b[k];
10 void mergesort(int a[],int l,int h){
11     if(l<h){
12         int m=(l+h)/2;
13         mergesort(a,l,m);
14         mergesort(a,m+1,h);
15         merge(a,l,m,h);
16 }
17 int main(){
18     int a[]={38,27,43,3,9,82,10},i,n=7;
19     mergesort(a,0,n-1);
20     for(i=0;i<n;i++)
21         printf("%d ",a[i]);
22     return 0;
23 }
```

Input

stdin input

Output

3 9 10 27 38 43 82